PlantCLEF 2026 — README SECTION

End-to-end inference pipeline for the **F1 = 0.50623** PlantCLEF 2026 submission (Exp-16C lock, 2026-05-05).

For a plain-English description of *what* each technique does and *why*, see `Working_Techniques_Summary.md`. This README covers *how to run the code*.

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1. Results

Milestone	Mechanism added	LB F1	Δ
Exp-3 baseline	noisy_or aggregator, T=2.0	0.28093	—
Exp-4	`max_prob` + JPEG q=85 + temporal `prior_blend`	0.35698	+0.07605
Exp-7/8	multiscale tiling 4+5+6 (77 tiles)	0.40100	+0.04402
Exp-9	ViT FAISS kNN blend (β_multi=0.80, β_single=1.00)	0.41754	+0.01654
Exp-11	ConvNeXt-L DINOv3 + ViT probability ensemble (w=0.77)	0.43790	+0.02036
Exp-14	per-source `sim_w` on RNNB + α=0.30	0.45410	+0.01620
Exp-14 Stage 12	aquatic-species habitat mask	0.45572	+0.00162
Exp-14 Stage 13	per-source biogeographic mask on alpine	0.45611	+0.00039
Exp-16 N3_r0.5	N-floor admission + ratio gate	0.46168	+0.00557
Exp-16	habitat-fit demotion (R=200 km, β=0.25)	0.49842	+0.03674
Exp-16C lock	**ViT-L+ConvNeXt cooldown backbones; per-source β override on alpine (β=0.45); habitat-fit retune (n_near_min=40, β_global=0.15)**	**0.50623**	**+0.00781**

Final submission CSV: `${PLANTCLEF_ROOT}/aerten3/submissions/exp16c/exp16c_s12B_alpine_b0.45.csv`

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2. Environment and prerequisites

- **Cluster**: Georgia Tech PACE.
- **SLURM account**: `paceship-clef2026_plantclef`.
- **GPU partitions**: `gpu-rtx6000` (default) or `gpu-h200` (training); QoS `inferno`.
- **CPU partitions**: `cpu-medium`; QoS `embers`.
- **Email for job notifications**: `aerten3@gatech.edu`.
- **Root path** (set before running anything):

```
export PLANTCLEF_ROOT=/storage/home/hcoda1/2/aerten3/ps-clef2026_plantclef-0
```

- **Python deps**: `torch`, `timm`, `faiss-cpu`, `pandas`, `pyarrow`, `numpy`, `Pillow`, `pyyaml`. The PACE conda environment used during the run is pinned in `Exp-10_ConvNext_finetuning/configs/`.
- **Data assumed present**:
  - Test images: `${PLANTCLEF_ROOT}/data/raw/PlanCLEF_test_data/images/`
  - Training images (1.4 M, used for FAISS index + habitat-fit lat/lon): `${PLANTCLEF_ROOT}/data/raw/train_full/`
  - Species map: `${PLANTCLEF_ROOT}/data/raw/class_id_to_species_mapping/class_to_species_id.json`
  - SW-Europe geo mask: `${PLANTCLEF_ROOT}/aerten3/species_masking/sw_europe_mask_bool.npy`

- PlantCLEF2024 training metadata for habitat-fit: `${PLANTCLEF_ROOT}/data/raw/PlantCLEF2024singleplanttrainingdata.csv`

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3. Repository layout

Folder	What it does	Status
`Exp-1_steps_with_lucas_212k_set/`	Original cross-region transformer + noisy-student attempt on the 212K LUCAS set	Abandoned (domain gap)
`Exp-2-JPEG_Compression/`	JPEG round-trip preprocessing	In pipeline
`Exp-4_Enhanced_Tile_Aggregation/`	`MaxProbabilityAggregator`, `prior_blend` temporal fusion	In pipeline
`Exp-7_Multiscale_tiling/`	Multi-scale tile extraction (ViT) at scales 4+5+6	In pipeline
`Exp-9_kNN_Retrieval_Ensemble/`	FAISS index build + per-tile kNN blend (ViT)	In pipeline
`Exp-10_ConvNext_finetuning/`	ConvNeXt-Large DINOv3 2-phase finetune + tile extraction	In pipeline
`Exp-10C_ConvNext_Cooldown/`	10-epoch cooldown of ConvNeXt-L on combined train+val+test	In pipeline (Exp-16C)
`Exp-11_Conv_ViT_Ensemble/`	ConvNeXt + ViT probability-level ensemble ("Approach A")	In pipeline
`Exp-13_ViT_finetuning/`	ViT-L DINOv2 2-phase finetune + Phase 2v3 cooldown	In pipeline (Exp-16C backbone)
`Exp-14_Adaptive_Temporal_Fusion/`	Per-source `sim_w` aggregator + aquatic + biogeographic masks	In pipeline
`Exp-16_Multi_Species_Discovery/`	N-floor admission + habitat-fit demotion	In pipeline
`Exp-16C_HabitatFit_Retune/`	Habitat-fit re-tune at the new ViT-L+ConvNeXt-cooldown backbones; per-source β override	**Final lock**
`Final_pipeline_methods_Summary/`	Detailed methods writeup (frozen at F1=0.4379 — superseded by Exp-16C)	Reference

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4. End-to-end run (final submission)

The minimal command sequence to reproduce **F1 = 0.50623** from raw test images and pre-trained checkpoints. Each line is a SLURM submission; later stages depend on earlier ones.

4.1 Train backbones (skip if checkpoints already exist)

```
# ConvNeXt-Large DINOv3, 2-phase finetune (4×H200, ~3 days total)
sbatch --export=VARIANT=large Exp-10_ConvNext_finetuning/slurm/phase1_train.sbatch
sbatch --export=VARIANT=large,GPU_TYPE=h200,NGPUS=4 Exp-
10_ConvNext_finetuning/slurm/phase2_train.sbatch

# ViT-L DINOv2, 2-phase + Phase 2v3 cooldown
sbatch Exp-13_ViT_finetuning/slurm/phase1_train.sbatch
sbatch Exp-13_ViT_finetuning/slurm/phase2_train.sbatch
sbatch Exp-13_ViT_finetuning/slurm/phase2v3_cooldown.sbatch

# ConvNeXt cooldown (Exp-10C) — same recipe applied to ConvNeXt
sbatch Exp-10C_ConvNext_Cooldown/slurm/cooldown.sbatch
```

4.2 Extract per-tile embeddings + logits at scales 4+5+6

```
# ViT-L tile extraction (3 scales × ~1.5 h on rtx6000)
for SCALE in 4 5 6; do
  sbatch --export=SCALE=$SCALE Exp-13_ViT_finetuning/slurm/extract_phase2v3_rtx6000.sbatch
done

# ConvNeXt-L cooldown tile extraction
sbatch Exp-10C_ConvNext_Cooldown/slurm/extract_4x4.sbatch
sbatch Exp-10C_ConvNext_Cooldown/slurm/extract_5x5_6x6.sbatch
```

Both extractors apply JPEG round-trip (q=85, 4:2:2) before forward pass and write parquet at `${PLANTCLEF_ROOT}/aerten3/parquet/.../scale_{4x4,5x5,6x6}.parquet`.

4.3 Build FAISS kNN index (ViT only, one-time)

```
sbatch Exp-9_kNN_Retrieval_Ensemble/slurm_build_index.sbatch
# Produces: faiss_index_geo.bin, faiss_labels_geo.npy, class_counts_geo.npy
```

4.4 ViT branch: kNN blend + temporal fusion

```
# Per-tile kNN blend (β_multi=0.80, β_single=1.00, k=20, τ=0.07)
# then sim_w aggregator on RNNB + max elsewhere, α_RNNB=0.30, α_default=0.15
sbatch Exp-14_Adaptive_Temporal_Fusion/slurm/run_stage4_simw_BOTH_vitRNNB30.sbatch
```

4.5 ConvNeXt branch: aggregation + temporal fusion

```
# max_prob over 77 tiles, T=1.5, image_topk=64, geo mask
# then sim_w on RNNB + max elsewhere, α=0.30
sbatch Exp-14_Adaptive_Temporal_Fusion/slurm/run_convnext_branch.sbatch
```

4.6 Probability-level ensemble + ecological masks

```
# w = 0.77 ViT+kNN + 0.23 ConvNeXt
# applies aquatic mask (Stage 12) and per-source biogeographic mask (Stage 13)
sbatch Exp-14_Adaptive_Temporal_Fusion/slurm/run_stage13_biogeo.sbatch
```

4.7 Habitat-fit demotion + per-source admission + per-source β (Exp-16C)

```
# Build per-species geographic profile (one-time, ~30 min CPU)
python Exp-16_Multi_Species_Discovery/diagnostics/build_species_geo_profile.py
python Exp-16_Multi_Species_Discovery/diagnostics/refine_source_centroids.py
python Exp-16_Multi_Species_Discovery/diagnostics/add_radius_columns.py

# Final lock: alpine β=0.45 override on top of global β=0.15
sbatch Exp-16C_HabitatFit_Retune/slurm/stage12B_alpine_beta_refine.sbatch
```

The final stage is implemented by `Exp-16_Multi_Species_Discovery/scripts/apply_habitat_fit.py`, invoked as:

```
python Exp-16_Multi_Species_Discovery/scripts/apply_habitat_fit.py \
  --ensemble-parquet  ${PLANTCLEF_ROOT}/aerten3/exp16c/ensemble_phase2v3_cnxt_cooldown.parquet \
  --refined-ref-json
${PLANTCLEF_ROOT}/aerten3/exp16/outputs/habitat_classifier/refined_source_ref.json \
```

```
  --species-geo-csv
${PLANTCLEF_ROOT}/aerten3/exp16/outputs/habitat_classifier/species_geo_profile.csv \
  --geo-mask          ${PLANTCLEF_ROOT}/aerten3/species_masking/sw_europe_mask_bool.npy \
  --aquatic-mask      ${PLANTCLEF_ROOT}/aerten3/exp14/outputs/signals/aquatic_mask.npy \
  --biogeo-json       ${PLANTCLEF_ROOT}/aerten3/exp14/outputs/signals/biogeo_per_source.npz \
  --radius-km 200 --beta 0.15 --n-near-min 40 --use-altitude \
  --per-source-json '{"CBN-PdlC":{"beta":0.45},"CBN-Pla":{"beta":0.45}}' \
  --n-floor 3 --ratio-floor 0.5 \
  --count-bound-sources GUARDEN,LISAH,OPTMix,RNNB \
  --threshold 0.085 --topk 10 \
  --out-csv ${PLANTCLEF_ROOT}/aerten3/submissions/exp16c/exp16c_s12B_alpine_b0.45.csv
```

This is the single command that produces the **F1 = 0.50623** submission CSV.

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5. Final hyperparameter manifest

```yaml
preprocessing:
  jpeg_quality: 85
  jpeg_subsampling: 1   # 4:2:2
  resize: lanczos

tiling:
  scales: [4, 5, 6]
  tiles_per_image: 77
  tile_size: 518

vit_branch:
  backbone: vit_large_patch14_reg4_dinov2.lvd142m  # Exp-13 phase2v3 cooldown checkpoint
  temperature: 1.5
  aggregator: max_prob
  image_topk: 64
  knn:
    index_type: IndexFlatIP
    k: 20
    tau: 0.07
    beta_multi: 0.80
    beta_single: 1.00
    blend_point: per_tile_pre_aggregation
  temporal_fusion:
    aggregator_per_source: {RNNB: sim_w, default: max}
    alpha_per_source: {RNNB: 0.30, default: 0.15}

convnext_branch:
  backbone: convnext_large.dinov3_lvd1689m  # Exp-10C cooldown checkpoint
  temperature: 1.5
  aggregator: max_prob
  image_topk: 64
  temporal_fusion:
    aggregator_per_source: {RNNB: sim_w, default: max}
    alpha: 0.30

ensemble:
  weights: {vit_plus_knn: 0.77, convnext: 0.23}

masks:
  geo: sw_europe_mask_bool.npy
  aquatic: 91 species (genus + epithet pattern match)
  biogeo_alpine: 5 species zeroed on CBN-PdlC, CBN-Pla

habitat_fit:
  beta_global: 0.15
  per_source_beta:
    CBN-PdlC: 0.45
```

```
    CBN-Pla:  0.45
  radius_km: 200
  n_near_min: 40
  use_altitude: true

admission:
  n_floor: 3
  ratio_floor: 0.5
  count_bound_sources: [GUARDEN, LISAH, OPTMix, RNNB]

submission:
  threshold: 0.085
  topk: 10
```

6. Key file index

Artefact	Path
Final submission CSV	`${PLANTCLEF_ROOT}/aerten3/submissions/exp16c/exp16c_s12B_alpine_b0.45.csv`
ViT-L Phase 2v3 cooldown checkpoint	`${PLANTCLEF_ROOT}/aerten3/checkpoints/exp13_vitl/phase2v3_cooldown/model_best.pth.tar`
ConvNeXt-L cooldown checkpoint	`${PLANTCLEF_ROOT}/aerten3/checkpoints/exp10c_convnext_cooldown/model_best.pth.tar`
FAISS index + labels	`${PLANTCLEF_ROOT}/aerten3/exp9_knn/faiss_index/{faiss_index_geo.bin, faiss_labels_geo.npy}`
Geo mask (SW Europe)	`${PLANTCLEF_ROOT}/aerten3/species_masking/sw_europe_mask_bool.npy`
Aquatic mask	`${PLANTCLEF_ROOT}/aerten3/exp14/outputs/signals/aquatic_mask.npy`
Per-source biogeographic mask	`${PLANTCLEF_ROOT}/aerten3/exp14/outputs/signals/biogeo_per_source.npz`
Refined source centroids	`${PLANTCLEF_ROOT}/aerten3/exp16/outputs/habitat_classifier/refined_source_ref.json`
Species geo profile	`${PLANTCLEF_ROOT}/aerten3/exp16/outputs/habitat_classifier/species_geo_profile.csv`
ViT tile parquets	`${PLANTCLEF_ROOT}/aerten3/parquet/plantclef_2026_test/exp13_phase2v3/scale_{4x4,5x5,6x6}.parquet`
ConvNeXt tile parquets	`${PLANTCLEF_ROOT}/aerten3/parquet/exp10c_convnext_cooldown/scale_{4x4,5x5,6x6}.parquet`